Exp: 7 Date:

Develop a Simple Calculator Using XMLRPC

Aim: To develop a simple calculator application Using XML-RPC Protocol in python that performs Basic arithmetic operations like addition, subraction, multiplication & division through client-server communication

Description: XML-RPC (extensible Markup Language Remote Procedure call) is a protocol that uses XML to encode requests & responses & HTTP as a transport mechanism. It enables communication between a client & a server where the client invokes methods hosted on the server remotely

* Server Holds arithmetic function
* The client connects to the server to perform operations like addition, subraction, multiplication & division

Code:

→ Server - Side code (server.py)

```
From Xmlrpc.server import
                    SimpleXLMRPC Server

def add (x,y):
    return x+y

def Subract (x,y):
    return x-y

def multiply (x,y):
    return x*y

def divide (x,y):
    if y!=0:
        return x/y
    else:
        return "Error: Division by zero"

Server = SimpleXMLRPC Server("localhost",
        8000, logRequests=False)

server.register_Function(add,"add")
server.register_Function(subract,"subract")
server.register_function(multiply,"multiply")
Server.register_Function(divide,"divide")
Print("XMLRPC Calculator Server is
        running on Port 8000...")
server.serve_forever()
```

```
def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Cannot divide by zero"
    return a / b
```

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```

num1, num2

Print("Inputs: 30 and 10")

Print("Addition (+) :", proxy.add(num1, num2))

Print("Subraction (-) :", proxy.subract(num1, num2))

Print("Multiplication (*) :", Proxy.multiply(num1, num2))

Print("Division (/) :", proxy.divide(num1, num2))

Software Requirements:

* OS: Window / Linux
* python Version: 3.X or above
* Modules required: xmlrpc.server, xmlrpc.client (built-in with python)
* Text Editor or IDE: IDLE / VS CODE / Pycharm

PROCEDURE:

1. Open a python IDE or Terminal
2. create two seperate python files: server.py and client.py
3. write the provided server-side code in server.py & client-side code in client.py
4. Run the server.py file first to start XML-RPC server
5. Run the client.py to perform arithmetic operations remotely
6. Observe the results displayed on client screen & verify the server output

OUTPUT: Server side

XML-RPC server is running on Port 8000...

client-side:-

Inputs: 30 and 10
Addition : 40
Subraction : 20
Multiplication: 300
Division : 3.0

Result: The XML-RPC Based calculator was succesfully implemented. The client communicated with the server using XML-RPC, performed all arithmetic operations remotely and displayed accurate results.